

Week -8

JOB SCHEDULING AND HUFFMAN CODING

JOB SCHEDULING:

Given:

The profits of the jobs are:

{36,21, 28, 19, 21, 13, 28, 25, 18, 20, 27, 22, 14, 26}

The deadlines of the jobs are:

{5, 8, 3, 4, 4, 9, 12, 14, 2, 7, 5, 1, 6, 3}

CODE:

```
#include<stdio.h>
struct Job{
    int id;
    int profit;
    int deadline;
};
void swap(struct Job *a,struct Job *b){
    struct Job t=*a;
    *a=*b;
    *b=t;
}
int main(){
    int profit[]={36,21,28,19,21,13,28,25,18,20,27,22,14,26};
    int deadline[]={5,8,3,4,4,9,12,14,2,7,5,1,6,3};
    int n=14;
    struct Job jobs[n];
    for(int i=0;i<n;i++){
        jobs[i].id=i+1;
        jobs[i].profit=profit[i];
        jobs[i].deadline=deadline[i];
    }
    for(int i=0;i<n;i++)
        for(int j=i+1;j<n;j++)
            if(jobs[i].profit<jobs[j].profit)
```

```

        swap(&jobs[i],&jobs[j]);
int slot[20]={0};
int total=0;
for(int i=0;i<n;i++){
    for(int j=jobs[i].deadline;j>0;j--){
        if(slot[j]==0){
            slot[j]=jobs[i].id;
            total+=jobs[i].profit;
            break;
        }
    }
}
printf("Optimal Job Sequence:\n");
for(int i=1;i<20;i++){
    if(slot[i])
        printf("Job%d ",slot[i]);
}
printf("\nTotal Profit=%d",total);
}

```

Optimal Job Sequence:

Job12 Job14 Job3 Job11 Job1 Job13 Job10 Job2 Job6 Job7 Job8
Total Profit=260

Job scheduling

Let there be $n=14$ jobs for sequencing whose profits are,

$P = \{ 36, 21, 28, 19, 21, 13, 28, 29, 18, 20, 27, 22, 34, 21 \}$
 $p = \{ 36, 28, 28, 27, 26, 24, 22, 21, 21, 20, 19, 18, 16, 13 \}$
 $D = \{ 5, 8, 3, 4, 11, 9, 12, 10, 2, 7, 5, 1, 6, 5 \}$

Step	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
[1st step]				J ₃	J ₈	J ₁			J ₂						
[2nd step]		J ₄	J ₃	J ₈	J ₁		J ₁₀	J ₂	J ₆			J ₇		J ₉	
[3rd step]	J ₁₂	J ₄	J ₃	J ₈	J ₁	J ₁₁	J ₁₀	J ₂	J ₆			J ₇		J ₉	

$\text{profit} = 27 + 18 + 28 + 21 + 36 + 14 + 20 + 21 + 13 + 28 =$
 $25 = 246 \text{ units}$

HUFFMAN CODING:

The string is “AMRITAVISHWAVIDYAPEETHAMCHENNAICAMPUS”

CODE:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define MAX 100

struct Node{
    char data;
    int freq;
    struct Node *left,*right;
};

struct Node* createNode(char data,int freq){
    struct Node* n=(struct Node*)malloc(sizeof(struct Node));
    n->data=data;
    n->freq=freq;
    n->left=n->right=NULL;
    return n;
}

void swap(struct Node **a,struct Node **b){
    struct Node *t=*a;
    *a=*b;
    *b=t;
}

void sort(struct Node *arr[],int n){
    for(int i=0;i<n;i++)
        for(int j=i+1;j<n;j++)
            if(arr[i]->freq > arr[j]->freq)
                swap(&arr[i],&arr[j]);
}
```

```
void printCodes(struct Node* root,int arr[],int top,int freq[],int *total){
```

```
    if(root->left){
        arr[top]=0;
        printCodes(root->left,arr,top+1,freq,total);
    }
```

```
    if(root->right){
        arr[top]=1;
        printCodes(root->right,arr,top+1,freq,total);
    }
```

```
    if(!root->left && !root->right){
```

```
        printf("%c : ",root->data);
```

```
        for(int i=0;i<top;i++)
            printf("%d",arr[i]);
```

```
        printf("\n");
```

```
        *total += freq[root->data]*top;
```

```
    }
}
```

```
int main(){
```

```
    char str[]="AMRITAVISHWAVIDYAPEETHAMCHENNAICAMPUS";
```

```
    int freq[256]={0};
```

```
    for(int i=0;str[i];i++)
        freq[str[i]]++;
```

```
    struct Node *nodes[MAX];
    int n=0;
```

```
    for(int i=0;i<256;i++){
```

```

    if(freq[i]>0){
        nodes[n++]=createNode(i,freq[i]);
    }
}

while(n>1){

    sort(nodes,n);

    struct Node *left=nodes[0];
    struct Node *right=nodes[1];

    struct Node *parent=createNode('$',left->freq+right->freq);
    parent->left=left;
    parent->right=right;

    nodes[0]=parent;

    for(int i=1;i<n-1;i++)
        nodes[i]=nodes[i+1];

    n--;
}

int arr[100],top=0,total=0;

printf("Huffman Codes:\n");

printCodes(nodes[0],arr,top,freq,&total);

printf("\nTotal Encoded Length = %d bits\n",total);

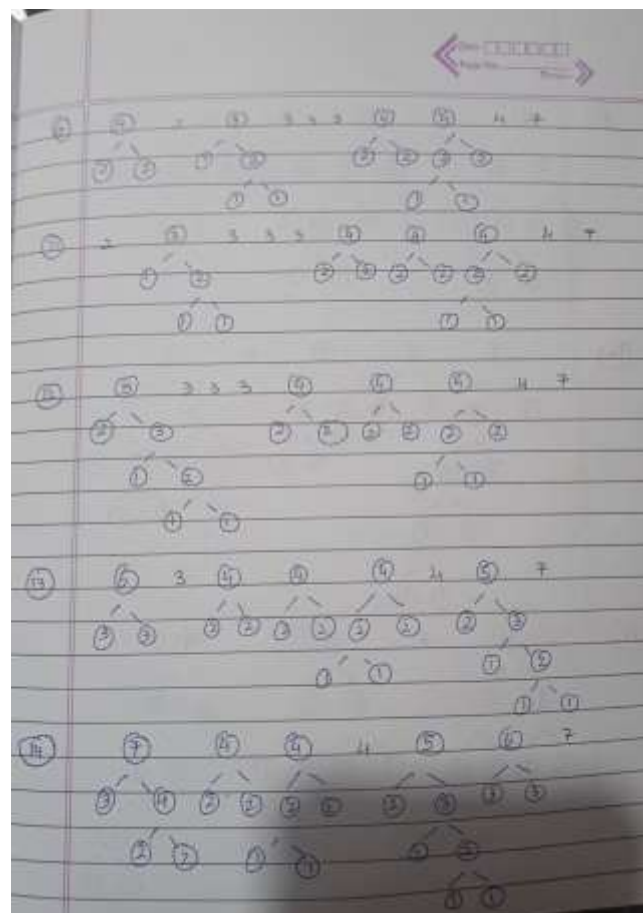
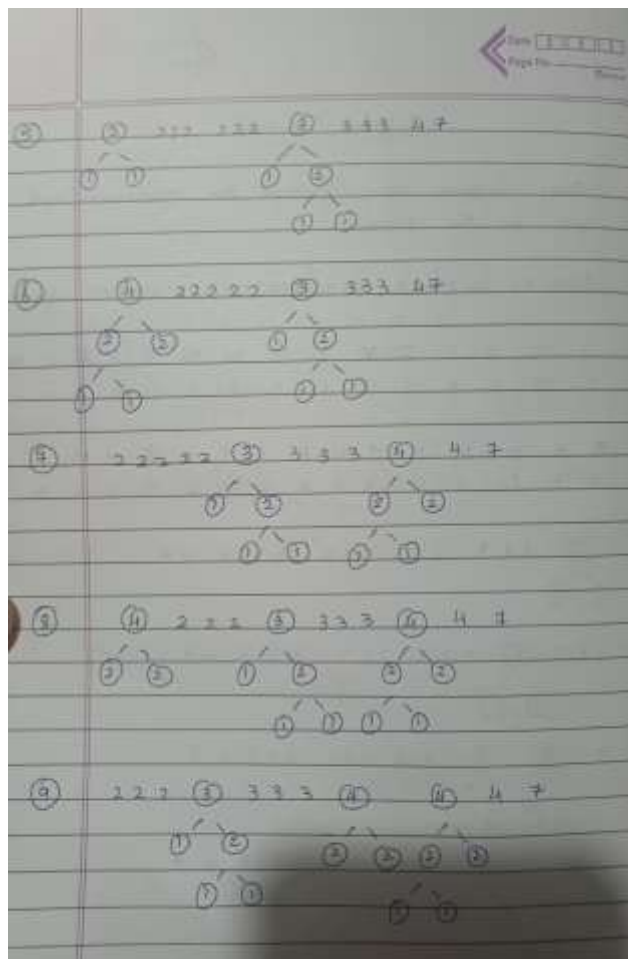
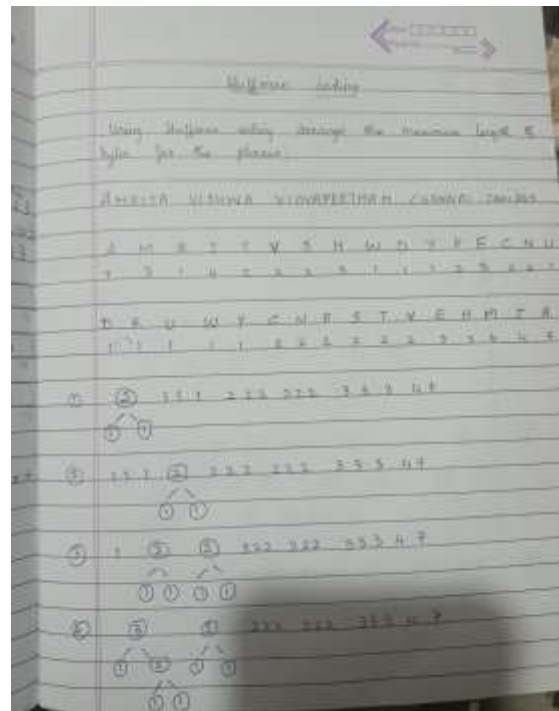
return 0;
}

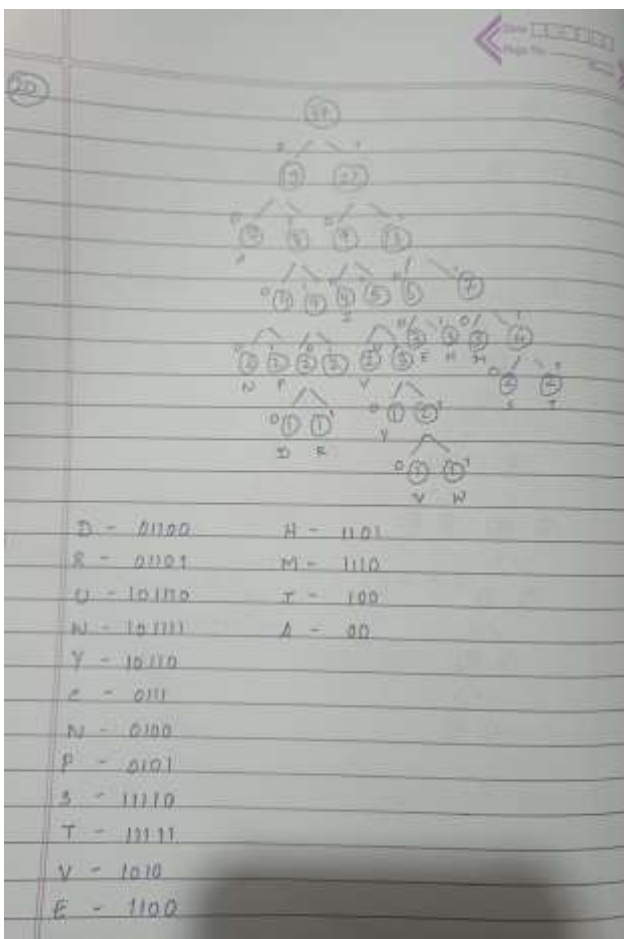
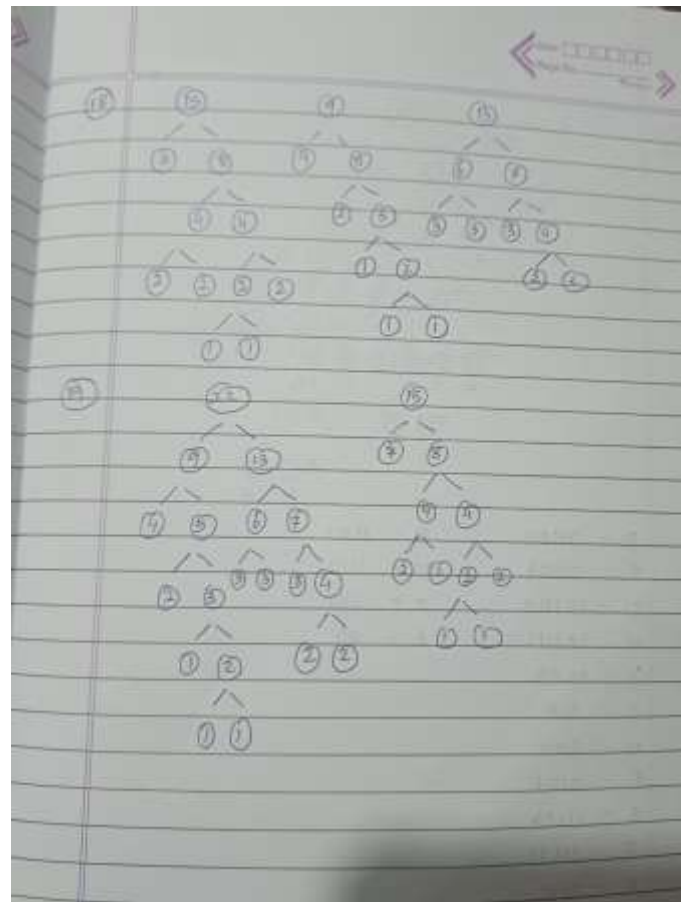
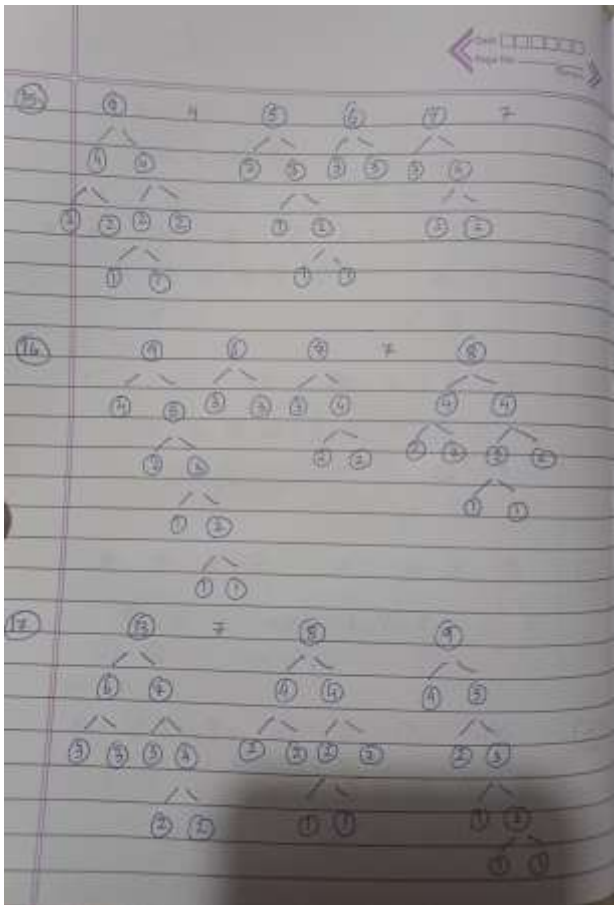
```

Huffman Codes:

A : 00
 T : 0100
 N : 0101
 D : 01100
 R : 01101
 S : 0111
 I : 100
 C : 1010
 Y : 10110
 U : 101110
 W : 101111
 E : 1100
 H : 1101
 M : 1110
 V : 11110
 P : 11111

Total Encoded Length = 141 bits





Handwritten calculations for Huffman coding:

$$\text{Avg. code} = \frac{5 + 9 + 8 + 6 + 8 + 8 + 8 + 10 + 10 + 10 + 16}{37}$$

$$\text{Avg. code} = \frac{114}{37}$$

$$= 3.08108$$

• Avg. code $\times$ 37

$$= 3.08108 \times 37$$

$$= 114 \text{ bits}$$